

VISHNU ASUTOSH DASU

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EDUCATION

- **Pennsylvania State University** *Aug 2024 - May 2028*
Doctor of Philosophy, Computer Science and Engineering
 - *Research Area:* Software Security, Program Analysis, Formal Verification, and Trustworthy AI
 - *Advisor:* Prof. Gang (Gary) Tan
- **Pennsylvania State University** *Aug 2022 - May 2024*
Master of Science, Computer Science and Engineering
 - *Thesis:* “Mitigating Unfairness in Deep Learning”
 - *Advisor:* Prof. Gang (Gary) Tan
- **Manipal Institute of Technology (MIT), Manipal** *July 2016 - July 2020*
Bachelor of Technology, Computer Science and Engineering

RESEARCH INTERESTS

I am a software security and trustworthy AI researcher. I develop automated analysis and repair techniques that improve the security, correctness, privacy, fairness, and robustness of software and AI systems. My work combines program analysis, formal verification, optimization, machine learning, and applied cryptography to build defenses with rigorous, provable guarantees.

Keywords: Software and systems security, trustworthy AI, program analysis and formal verification, privacy-preserving machine learning, applied cryptography

SELECTED PUBLICATIONS

* - Equal Contribution, † - Alphabetical Order

- **Attention Pruning: Automated Fairness Repair of Language Models via Surrogate Simulated Annealing** [pdf]
48th International Conference on Software Engineering (ICSE), 2026
Vishnu Asutosh Dasu, Md Rafi ur Rashid, Vipul Gupta, Saeid Tizpaz-Niari, Gang Tan
- **Improving Noise Efficiency in Privacy-preserving Dataset Distillation** [pdf]
International Conference on Computer Vision (ICCV), 2025
Runkai Zheng, Vishnu Asutosh Dasu, Yinong Wang, Haohan Wang, Fernando De la Torre
- **Privacy-Preserving Data Deduplication for Enhancing Federated Learning of Language Models** [pdf]
Network and Distributed System Security (NDSS) Symposium, 2025
Aydin Abadi*, Vishnu Asutosh Dasu*, Sumanta Sarkar*
- **Chain-of-Thought Driven Adversarial Scenario Extrapolation for Robust Language Models** [pdf]
40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026
Md Rafi Ur Rashid, Vishnu Asutosh Dasu, Ye Wang, Gang Tan, Shagufta Mehnaz
- **NeuFair: Neural Network Fairness Repair with Dropout** [pdf]
33rd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), 2024
Vishnu Asutosh Dasu, Ashish Kumar, Saeid Tizpaz-Niari, Gang Tan
- **Gradient-Free Privacy Leakage in Federated Language Models through Selective Weight Tampering** [pdf]
Privacy Enhancing Technologies Symposium (PETS) 2026

Md Rafi Ur Rashid, **Vishnu Asutosh Dasu**, Kang Gu, Najrin Sultana, Shagufta Mehnaz

- **PROV-FL: Privacy-preserving Round Optimal Verifiable Federated Learning [pdf]**
15th ACM Workshop on Artificial Intelligence and Security (AISEC), ACM CCS, 2022
Vishnu Asutosh Dasu, Sumanta Sarkar, Kalikinkar Mandal
- **Side Channel Attack On Stream Ciphers: A Three-Step Approach To State/Key Recovery [pdf]**
IACR Conference on Cryptographic Hardware and Embedded Systems (CHES), 2022
Satyam Kumar, **Vishnu Asutosh Dasu**, Anubhab Baksi, Santanu Sarkar, Dirmanto Jap, Jakub Breier, Shivam Bhasin
- **Impact of Data Duplication on Deep Neural Network-Based Image Classifiers: Robust vs. Standard Models [pdf]**
IEEE Deep Learning Security and Privacy Workshop (DLSP), IEEE S&P, 2025
Alireza Aghabagherloo, Aydin Abadi, Sumanta Sarkar, **Vishnu Asutosh Dasu**, Bart Preneel
- **EvoquerBot: A multimedia chatbot leveraging synthetic data for cross-domain assistance [pdf]**
Alexa Prize TaskBot Challenge 2 Proceedings
Team EvoquerBOT, Penn State University
- **New Results on Machine Learning-Based Distinguishers [pdf]**
IEEE Access, 2023
Anubhab Baksi†, Jakub Breier†, **Vishnu Asutosh Dasu**†, Xiaolu Hou†, Hyunji Kim†, Hwajeong Seo†
- **Three Input Exclusive-OR Gate Support For Boyar-Peralta’s Algorithm [pdf]**
22nd International Conference on Cryptology in India (Indocrypt), 2021
Anubhab Baksi, **Vishnu Asutosh Dasu**, Banashri Karmakar, Anupam Chattopadhyay, Takanori Isobe
- **LIGHTER-R: Optimized Reversible Circuit Implementation For SBoxes**
32nd IEEE International System-on-Chip Conference (SOCC), 2019
Vishnu Asutosh Dasu, Anubhab Baksi, Sumanta Sarkar, Anupam Chattopadhyay
- **[Re] GANSpace: Discovering Interpretable GAN Controls**
ReScience C, 2022
Vishnu Asutosh Dasu, Midhush Manohar T.K.

PREPRINTS/UNDER REVIEW

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- **Heimdall: Formally Verified Automated Migration of Legacy eBPF Programs to Rust [pdf]**
(Under Submission) USENIX Security 2027
Vishnu Asutosh Dasu, Monika Santra, Md Rafi Ur Rashid, Ashish Kumar, Saeid Tizpaz-Niari, Gang Tan
 - **Robustness of Vision Language Models Against Split-Image Harmful Input Attacks [pdf]**
(Under Review) ACM Conference on Computer and Communications Security (CCS) 2026
Md Rafi Ur Rashid, Md Sadik Hossain Shanto, **Vishnu Asutosh Dasu**, Shagufta Mehnaz
 - **iResolveX: Multi-Layered Indirect Call Resolution via Static Reasoning and Learning-Augmented Refinement [pdf]**
(Under Submission) Network and Distributed System Security (NDSS) Symposium, 2027
Monika Santra, Bokai Zhang, Mark Lim, **Vishnu Asutosh Dasu**, Dongrui Zeng, Gang Tan
 - **Finding Probabilistic Sub-invariants via Search Space Algorithms under Noisy Cost Functions**

(Under Review) Formal Methods in Computer-Aided Design (2026)
Ashish Kumar, **Vishnu Asutosh Dasu**, Saeid Tizpaz-Niari, Gang Tan

PATENTS

- **A Method to Secure Dynamic Agentic Artificial Intelligence Systems from Prompt Injection Attacks**
(Filing in Progress) 2026
Vishnu Asutosh Dasu, Lilit Avetisyan, Jorge Guajardo Merchan

ACADEMIC AND WORK EXPERIENCE

- **Bosch Center for AI** *May 2026 - Aug. 2026*
AI Research Intern *Advisor: Jorge Guajardo Merchan*
 - *Projects:* System-level Defenses for Securing AI Agents
- **Tata Research Development and Design Centre (TRDDC)** *Sept 2020 - June 2022*
Researcher, Cybersecurity and Privacy *Advisor: Sumanta Sarkar*
 - *Projects:* Privacy-preserving Federated Learning, Secure Multiparty Computation, Insider Threat Detection
- **Citrix R&D** *Jan 2020 - June 2020*
Software Engineer Intern, Citrix Analytics for Security (CAS)
 - Full-stack Web Developer (Spring Boot, React.js)
- **Nanyang Technological University (NTU)** *Dec 2019 - Jan 2020*
Research Intern *Advisor: Anupam Chattopadhyay*
 - *Projects:* Optimizing hardware implementations of block ciphers
- **Tata Consultancy Services (TCS) Research** *May 2019 - July 2019*
Research Intern, Cybersecurity and Privacy *Advisor: Chalamala Srinivasa Rao*
 - *Projects:* Adversarial attacks and defenses on Convolutional Neural Networks (CNNs)
- **Indian Statistical Institute** *May 2018 - July 2018*
Summer Scholar *Advisor: Dipti Prasad Mukherjee*
 - *Project:* 3-D coordinate estimation of humans from 2-D live video feed

SKILLS

- **Languages:** C/C++, Rust, Python, Java, Go, SQL
- **Systems & Formal Methods:** LLVM, Clang, eBPF (libbpf, Aya), Z3/SMT solvers, Alive2, angr, KLEE, Lean, symbolic execution
- **AI/ML:** Large Language Models (LLMs) and agentic pipelines, PyTorch, Trustworthy AI (Fairness, Privacy, Robustness)
- **Areas of Expertise:** Software Security, Program Analysis, Trustworthy AI, Applied Cryptography

SERVICE

- **Program Committee:** USENIX Security 2027, AAAI 2026
- **Reviewer:** IEEE TDSC (2026), IEEE TIFS (2025), IEEE Access (2025)
- **Artifact Evaluation Committee:** ASE 2026, ISSTA 2026, ACM CCS 2026, USENIX Security 2026, ACM CCS 2025

- **External Reviewer:** NDSS 2025, OOPSLA 2025, IEEE S&P 2025, IEEE S&P 2026
- **Organizer:** Penn State Security Reading Group (2024 - Present)
- **Judge:** Penn State Undergraduate Exhibition 2025

AWARDS AND ACHIEVEMENTS

- **Future Leader of AI:** Selected as a Future Leader of AI at the inaugural *ACM AI Leadership Summit*, 2026 with a scholarship.
- **Scholarship:** Received a scholarship to attend the *Fifteenth Summer School on Formal Techniques* in Menlo Park, CA.
- **Student Travel Grant:** Received the ACM/SIGSOFT CAPS travel grant to attend ICSE 2026.
- **Distinguished Reviewer Award:** Received the distinguished reviewer award for artifact evaluation in ACM CCS 2025.
- **Student Travel Grant:** Received the travel grant to attend ICCV 2025.
- **Internet Society Fellowship:** Received the Internet Society Fellowship to attend NDSS 2025.
- **TCS Citation Award** ($3 \times$ recipient): Received the award for outstanding research contributions to Tata Consultancy Services (TCS).
- **Scholarship:** Received a scholarship to attend the *Winter School on Responsible AI* in Israel.
- **Best Project Award:** Received the Best Project Award during the *Fifth Summer School on Computer Vision, Graphics and Image Processing*, Indian Statistical Institute (ISI) Kolkata.
- **IGVC:** Placed 2nd in the Interoperability Profiles Challenge and 9th overall at *Intelligent Ground Vehicle Competition (IGVC)* 2018. Second-best among all teams from India.
- **ACM ICPC Regionals:** Represented MIT Manipal at the 2017 *ACM ICPC Asia Regional Contest*.
- **DAGsHub Award:** Received a \$500 award from *DAGsHub* for completing the *ML Reproducibility Challenge Spring 2021*.

GRANTS

- **NSF NAIRR 260079:** Automated and Verified Migration of Legacy eBPF Programs to Rust. Total Award: \$8,000. 2026–2027.

PROFESSIONAL MEMBERSHIPS

- Student Member of the ACM
- Student Member of IEEE

REFERENCES

- Prof. Gang Tan (Advisor): gtan@psu.edu
- Prof. Saeid Tizpaz-Niari: saeid@uic.edu
- Prof. Sumanta Sarkar: sumanta.sarkar@essex.ac.uk